Car Showroom Sales Analysis & Price Prediction

Comprehensive End-to-End Data Science Project Report | Automobile Retail Operations

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Domain: Automotive Intelligence

Records Analyzed: 600 Vehicles

Status: Complete & Verified

1. Executive Summary & Core Business Objectives

This study examines vehicle sales transactions across 600 retail units to determine demand patterns, evaluate sales force throughput, identify high-profit automobile segments, and build an automated pricing prediction engine. Showroom revenue totaled **₹1,007,326,572** with a cumulative gross profit of **₹129,871,508**. Toyota, Honda, and Maruti Suzuki generated the highest unit volumes, while SUVs yielded the single largest category profit contribution. Three regression algorithms were benchmarked for automated trade-in valuation, with Linear Regression achieving a winning predictive accuracy of **R^2 = 0.9923**.

2. High-Level Financial & Operational KPIs

TOTAL UNITS SOLD	TOTAL GROSS REVENUE	TOTAL NET PROFIT	AVG PURCHASE COST
600	**₹1.007 B**	**₹129.87 M**	**₹1,462,425**
100% Verified Sales	Avg ₹1,678,877 / Car	12.89% Net Showroom Margin	Spread: ₹216,452 / Unit

3. Data Collection, Hygiene & Attribute Verification

The dataset was ingested using Python (Pandas) across 18 specialized operational columns covering vehicle specifications, customer demographics, financial transactions, and sales allocation. Strict cleaning protocols confirmed zero missing values and zero duplicate records. Outliers in vehicle mileage and selling prices were audited using the Interquartile Range (IQR) method; observed kilometers (5,000–120,000 km) and prices (₹450k–₹2.99M) fell strictly inside statistical tolerance thresholds.

Data Field	Type	Null / Dups	Description & Distribution Range
Salesperson / City / Mode	Categorical	0 / 0	4 sales executives; 5 regional hubs; 4 payment modes.

4. Exploratory Data Analysis & Business Intelligence

Detailed aggregation of showroom transactions isolates specific consumer purchasing patterns, sales associate performance, and vehicle profitability. The showroom operates with balanced sales distribution across quarters, but experiences noticeable mid-year volatility requiring inventory calibration.

Top Vehicle Brands by Sales Volume

Brand Name	Units Sold	Market Share
Toyota	**110**	**18.33%**
Honda	106	17.67%
Maruti Suzuki	103	17.17%
Hyundai	102	17.00%
Mahindra	97	16.17%
Tata	82	13.67%

Top Models by Customer Demand

Model	Brand	Units Sold
Innova	**Toyota**	**45**
Creta	Hyundai	42
Amaze	Honda	40
Thar	Mahindra	37
Baleno	Maruti Suzuki	36
Harrier	Tata	32

Category Profitability Matrix

SUVs generated the largest total profit (**₹58,865,158**), followed closely by Hatchbacks (**₹58,028,048**). Sedans produced lower volume and aggregate profit (**₹12,978,302**).

Category	Cumulative Profit (₹)	Share
SUV	**₹58,865,158**	**45.33%**
Hatchback	₹58,028,048	44.68%
Sedan	₹12,978,302	9.99%

Sales Associate Revenue Performance

Throughput across all four sales executives is highly competitive, led by Sneha Patil with ₹266.9M in revenue, followed closely by Rahul Sharma.

Salesperson	Total Revenue (₹)	Contribution
Sneha Patil	**₹266,930,545**	**26.50%**
Rahul Sharma	₹260,771,833	25.89%
Priya Kulkarni	₹249,457,331	24.76%
Amit Deshmukh	₹230,166,863	22.85%

5. Fuel Type & Powertrain Dynamics

Analysis of powertrain selection indicates that conventional internal combustion remains dominant: **Petrol accounts for 50.3%** (302 units), followed by **Diesel at 29.8%** (179 units), **CNG at 16.3%** (98 units), and **Electric at 3.5%** (21 units). Manual transmissions comprise 64.2% of transactions, driven by budget-conscious hatchback buyers, while Automatics represent 35.8%, clustering in mid-to-high trim SUVs and premium sedans.

6. Comprehensive Visual Analytics Gallery

The 6-panel analytical visualization below encapsulates the core structural characteristics of the car showroom's operational performance, seasonal sales fluctuations, customer powertrain adoption, and brand-level price dispersion.



Figure 1: Multi-panel exploratory visualization generated via Seaborn & Matplotlib from the verified 600-vehicle inventory dataset.

Detailed Chart Interpretations & Strategic Findings

- **Plot 1 (Sales by Brand):** Toyota (110 units) and Honda (106 units) demonstrate the strongest showroom turnover, underscoring enduring resale confidence in Japanese engineering.
- **Plot 2 (Monthly Sales Trend):** Sales reach an annual peak in *June (66 units)*, followed by an immediate sharp monsoon contraction in *August (35 units)* before rebounding during festival stocking in September (53 units).
- **Plot 3 (Fuel Distribution):** Petrol retains an absolute majority (50.3%), with Diesel preserving significant commercial demand (29.8%). Electric vehicles (3.5%) represent a fledgling but high-growth niche.
- **Plot 4 (Selling Price Distribution):** Prices show a healthy bimodal distribution peaking between ₹800K–₹1.2M (entry hatchbacks) and ₹2.2M–₹2.6M (premium SUVs and MPVs).
- **Plot 5 (Kilometers vs. Price):** Selling price is strongly anchored to initial purchase valuation, with mileage exerting smooth linear depreciation without catastrophic structural cliffs.
- **Plot 6 (Price Range by Brand):** Mahindra, Toyota, and Hyundai exhibit the widest interquartile spreads due to diversified model ranges extending from mass-market utility to luxury flagships.

7. Machine Learning Model Development & Benchmark

A supervised regression pipeline was constructed to predict **Selling_Price** based on physical specifications and acquisition economics. The 600 records were partitioned into an 80% training set (480 units) and a 20% holdout test set (120 units). Categorical features (`Brand`, `Fuel_Type`, `Transmission`) were encoded using One-Hot Encoding with first-tier drop to avoid dummy collinearity, while continuous numeric attributes (`Manufacturing_Year`, `Kilometers_Driven`, `Purchase_Price`) were passed through unified pipelines.

Evaluated Algorithm	MAE (₹)	MSE	RMSE (₹)	R^2 Score	Model Status
Linear Regression	**₹48,212.26**	**3,851,326,505.73**	**₹62,059.06**	**0.9923**	**Recommended for Deployment**
Random Forest Regressor (100 Trees)	₹51,079.18	4,519,761,382.85	₹67,229.17	0.9909	Benchmarked Alternative
Decision Tree Regressor (Depth = 8)	₹63,825.79	7,277,004,672.14	₹85,305.36	0.9854	Baseline Tree Model

Algorithmic Interpretation & Selection Justification

Linear Regression outperformed tree-based ensembles across every single quantitative metric, achieving the lowest Mean Absolute Error (**₹48,212.26**) and highest coefficient of determination (**R^2 = 0.9923**). Because used vehicle pricing margins are largely calculated as a steady markup over acquisition cost with systematic depreciation per age and mileage, the underlying mechanics possess strong linear relationships. Linear Regression offers instant inference latency, total interpretability, and zero risk of leaf-node overfitting.

8. Strategic Business Recommendations & Next Steps

1. Sourcing & Stock Optimization

Reallocate working capital to secure steady inventory of Toyota Innova and Hyundai Creta. SUVs and Hatchbacks drive 90.0% of all gross profits; maintain minimum 45-day floor inventory for both segments.

2. Monsoon Promotion Smoothing

Deploy promotional cash discounts, exchange bonuses, and extended warranty bundles during July and August to overcome the severe mid-year sales drop-off (down 47% from June peak).

3. Automated Valuation Tool Deployment

Integrate the trained Linear Regression model into the showroom tablet CRM. Sales representatives can generate instant, unbiased purchase and resale quotes for incoming customer trade-ins.

4. Sales Force Incentive Alignment

Implement targeted commission tiers incentivizing the movement of high-margin SUV inventory and slow-moving sedans, maintaining balanced gross revenue generation across all 4 associates.

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Project Lead Review: Approved / Verified

Artifacts: Notebook (.ipynb), Datasets (.csv), Report (.pdf)